

# COMMERCIAL SOLAR — FINANCIAL ANALYSIS

Returns, plant and tax depreciation

500 kWp

PLANT CAPACITY

APSPDCL · COMMERCIAL ROOFTOP

## 500 KWP GRID-TIED SOLAR PLANT

Net investment ₹1.93 Cr after input tax credit, generating 7,66,500 units in year one against an energy charge of ₹7.65 per unit.

HT CATEGORY I / II AT 11 KV

30-YEAR ANALYSIS

PAYBACK

3.7 yrs

Post-tax, undiscounted

PROJECT IRR

27.5%

Unlevered, 30 years

COST OF SOLAR ENERGY

₹3.15

per kWh, against ₹7.65 from the grid

NET PRESENT VALUE

₹3.56 Cr

Discounted at 10%

### PLANT AND GENERATION

|                      |                                                  |
|----------------------|--------------------------------------------------|
| Plant capacity       | 500 kWp                                          |
| Year 1 generation    | 7,66,500 kWh                                     |
| Specific yield       | 1,533 kWh / kWp / year                           |
| Supply category      | HT Category I / II at 11 kV (HT supply at 11 kV) |
| Energy charge offset | ₹7.65 / kWh                                      |
| Units self-consumed  | 90% at the energy charge                         |
| Units exported       | 10% at ₹2.75 / kWh (APPC)                        |
| Year 1 gross saving  | ₹54.88 L                                         |

### INVESTMENT AND TAX BASIS

|                              |                                |
|------------------------------|--------------------------------|
| Price per watt-peak          | ₹40.50 / Wp (incl. GST)        |
| Project price incl. GST      | ₹2.02 Cr                       |
| GST at 5% — recovered as ITC | ₹9.64 L                        |
| Net capitalised cost         | ₹1.93 Cr                       |
| Depreciation block           | 40% WDV, Appendix I            |
| Tax regime                   | s.115BAA — 25.168%             |
| Annual maintenance           | ₹600 / kWp / year, year 1 free |
| 30-year depreciation shield  | ₹48.54 L                       |

### ACCELERATED DEPRECIATION — FIRST 12 YEARS

| YEAR                   | OPENING WDV (₹) | DEPRECIATION AT 40% (₹) | CLOSING WDV (₹) | TAX SHIELD AT 25.168% (₹) |
|------------------------|-----------------|-------------------------|-----------------|---------------------------|
| 1                      | 1,92,85,715     | 77,14,286               | 1,15,71,429     | 19,41,532                 |
| 2                      | 1,15,71,428     | 46,28,571               | 69,42,857       | 11,64,919                 |
| 3                      | 69,42,857       | 27,77,143               | 41,65,714       | 6,98,951                  |
| 4                      | 41,65,715       | 16,66,286               | 24,99,429       | 4,19,371                  |
| 5                      | 24,99,428       | 9,99,771                | 14,99,657       | 2,51,622                  |
| 6                      | 14,99,657       | 5,99,863                | 8,99,794        | 1,50,974                  |
| 7                      | 8,99,795        | 3,59,918                | 5,39,877        | 90,584                    |
| 8                      | 5,39,877        | 2,15,951                | 3,23,926        | 54,351                    |
| 9                      | 3,23,926        | 1,29,570                | 1,94,356        | 32,610                    |
| 10                     | 1,94,355        | 77,742                  | 1,16,613        | 19,566                    |
| 11                     | 1,16,613        | 46,645                  | 69,968          | 11,740                    |
| 12                     | 69,968          | 27,987                  | 41,981          | 7,044                     |
| Tax shield, years 1-12 |                 |                         |                 | 48,43,263                 |

# COMMERCIAL SOLAR — FINANCIAL ANALYSIS

30-year post-tax cash flow

**500 kWp**

PLANT CAPACITY

**How this table works.** Electricity savings reduce an operating expense, so they increase taxable profit and are taxed at 25.168%. Depreciation reduces the same taxable profit. Both are applied to one figure, so the tax column is a net inflow in the early years when depreciation exceeds savings — shown in green. Net cash flow is therefore gross saving, less maintenance, less that tax. The shaded row is the year the investment is recovered.

| YEAR         | GENERATION (KWH)   | GROSS SAVING (₹)    | AMC (₹)          | DEPRECIATION (₹)   | TAX (₹)            | NET CASH FLOW (₹)   | CUMULATIVE (₹)      |
|--------------|--------------------|---------------------|------------------|--------------------|--------------------|---------------------|---------------------|
| 1            | 7,66,500           | 54,88,140           | —                | 77,14,286          | (5,60,276)         | 60,48,416           | 60,48,416           |
| 2            | 7,62,668           | 56,79,127           | 3,00,000         | 46,28,571          | 1,88,900           | 51,90,227           | 1,12,38,643         |
| 3            | 7,58,854           | 58,76,761           | 3,00,000         | 27,77,143          | 7,04,608           | 48,72,153           | 1,61,10,796         |
| 4            | 7,55,060           | 60,81,272           | 3,00,000         | 16,66,286          | 10,35,660          | 47,45,612           | 2,08,56,408         |
| 5            | 7,51,285           | 62,92,900           | 3,00,000         | 9,99,771           | 12,56,671          | 47,36,230           | 2,55,92,638         |
| 6            | 7,47,528           | 65,11,893           | 3,00,000         | 5,99,863           | 14,12,436          | 47,99,458           | 3,03,92,096         |
| 7            | 7,43,791           | 67,38,507           | 3,00,000         | 3,59,918           | 15,29,859          | 49,08,648           | 3,53,00,744         |
| 8            | 7,40,072           | 69,73,007           | 3,00,000         | 2,15,951           | 16,25,112          | 50,47,895           | 4,03,48,639         |
| 9            | 7,36,371           | 72,15,668           | 3,00,000         | 1,29,570           | 17,07,925          | 52,07,743           | 4,55,56,382         |
| 10           | 7,32,689           | 74,66,773           | 3,00,000         | 77,742             | 17,84,167          | 53,82,606           | 5,09,38,988         |
| 11           | 7,29,026           | 77,26,617           | 3,00,000         | 46,645             | 18,57,391          | 55,69,226           | 5,65,08,214         |
| 12           | 7,25,381           | 79,95,503           | 3,00,000         | 27,987             | 19,29,760          | 57,65,743           | 6,22,73,957         |
| 13           | 7,21,754           | 82,73,747           | 3,00,000         | 16,792             | 20,02,606          | 59,71,140           | 6,82,45,097         |
| 14           | 7,18,145           | 85,61,673           | 3,00,000         | 10,075             | 20,76,762          | 61,84,911           | 7,44,30,008         |
| 15           | 7,14,554           | 88,59,619           | 3,00,000         | 6,045              | 21,52,764          | 64,06,856           | 8,08,36,864         |
| 16           | 7,10,982           | 91,67,934           | 3,00,000         | 3,627              | 22,30,969          | 66,36,965           | 8,74,73,829         |
| 17           | 7,07,427           | 94,86,978           | 3,00,000         | 2,176              | 23,11,631          | 68,75,347           | 9,43,49,176         |
| 18           | 7,03,890           | 98,17,125           | 3,00,000         | 1,306              | 23,94,941          | 71,22,184           | 10,14,71,360        |
| 19           | 7,00,370           | 1,01,58,761         | 3,00,000         | 783                | 24,81,056          | 73,77,705           | 10,88,49,065        |
| 20           | 6,96,868           | 1,05,12,286         | 3,00,000         | 470                | 25,70,110          | 76,42,176           | 11,64,91,241        |
| 21           | 6,93,384           | 1,08,78,113         | 3,00,000         | 282                | 26,62,229          | 79,15,885           | 12,44,07,126        |
| 22           | 6,89,917           | 1,12,56,672         | 3,00,000         | 169                | 27,57,533          | 81,99,139           | 13,26,06,265        |
| 23           | 6,86,467           | 1,16,48,404         | 3,00,000         | 102                | 28,56,141          | 84,92,263           | 14,10,98,528        |
| 24           | 6,83,035           | 1,20,53,768         | 3,00,000         | 61                 | 29,58,173          | 87,95,595           | 14,98,94,123        |
| 25           | 6,79,620           | 1,24,73,240         | 3,00,000         | 37                 | 30,63,752          | 91,09,488           | 15,90,03,611        |
| 26           | 6,76,222           | 1,29,07,308         | 3,00,000         | 22                 | 31,73,002          | 94,34,306           | 16,84,37,917        |
| 27           | 6,72,841           | 1,33,56,483         | 3,00,000         | 13                 | 32,86,052          | 97,70,430           | 17,82,08,347        |
| 28           | 6,69,477           | 1,38,21,288         | 3,00,000         | 8                  | 34,03,036          | 1,01,18,252         | 18,83,26,599        |
| 29           | 6,66,129           | 1,43,02,269         | 3,00,000         | 5                  | 35,24,090          | 1,04,78,179         | 19,88,04,778        |
| 30           | 6,62,798           | 1,47,99,988         | 3,00,000         | 3                  | 36,49,356          | 1,08,50,632         | 20,96,55,410        |
| <b>Total</b> | <b>2,14,03,105</b> | <b>28,23,81,824</b> | <b>87,00,000</b> | <b>1,92,85,709</b> | <b>6,40,26,416</b> | <b>20,96,55,410</b> | <b>20,96,55,410</b> |

RETURN METRICS

|                          |          |
|--------------------------|----------|
| Simple payback           | 3.7 yrs  |
| Discounted payback       | 4.9 yrs  |
| Project IRR, unlevered   | 27.5%    |
| Net present value at 10% | ₹3.56 Cr |

OVER THE FULL TERM

|                                |                 |
|--------------------------------|-----------------|
| Levelised cost of energy       | ₹3.15 / kWh     |
| Cumulative return, 30 years    | 987%            |
| Total net cash flow, after tax | ₹20.97 Cr       |
| 30-year generation             | 2,14,03,103 kWh |

COST OF A UNIT — SOLAR AGAINST THE GRID

|                                |  |       |
|--------------------------------|--|-------|
| Solar, levelised over 30 years |  | ₹3.15 |
| Grid energy charge today       |  | ₹7.65 |

Every unit the plant generates costs ₹3.15 across its life against ₹7.65 bought from APSPDCL today — and the grid figure reaches ₹23.86 by year 30 at 4% escalation, while the solar figure is fixed at the day of purchase. Over 30 years the plant generates 2,14,03,103 units, avoiding 17,550.5 tonnes of CO<sub>2</sub>.

ASSUMPTIONS

- Tariff:** HT Category I / II at 11 kV at ₹7.65/kWh. Energy charge; ₹475/kVA/month demand charge excluded. APERC Retail Supply Tariff Order FY2025-26, continued for FY2026-27.
- Demand charges:** Excluded. Solar does not reduce contracted demand, so fixed and demand charges are unchanged by the plant.
- Tariff escalation:** 4% per year, applied to both the energy charge and the export rate.
- Generation:** 1,533 kWh per kWp in year one, degrading 0.5% per year.
- Unit split:** 90% self-consumed, 10% exported at ₹2.75/kWh. The plant is assumed to be sized to the site's daytime load.
- GST:** 5%, recovered in full as input tax credit. Figures are stated ex-GST. A buyer who cannot claim ITC should read the cost as ₹2.02 Cr.
- Depreciation:** 40% written-down value, Income Tax Act Appendix I. No additional depreciation under s.32(1)(ia), which s.115BAA forfeits.
- Tax:** 25.168% — s.115BAA at 22% plus 10% surcharge and 4% cess. Applied to savings and to depreciation alike.
- Loss set-off:** Early-year depreciation exceeds savings. The resulting loss is assumed to shelter the buyer's other business income in the same year.
- Commissioning:** Full first-year depreciation assumed, i.e. the plant is put to use on or before 30 September. Later commissioning halves the year-one charge, with the balance carried forward.
- Maintenance:** ₹600 per kWp per year from year 2, held flat. Year 1 is covered by the build.
- Inverter replacement:** Not modelled. A mid-life inverter replacement would reduce the figures shown.
- Financing:** None. All returns are unlevered — the return on the money invested, before any loan.
- Horizon:** 30 years, which runs beyond the module performance warranty.
- Discount rate:** 10% for net present value, discounted payback and levelised cost.
- Emissions:** Grid emission factor 0.82 kg CO<sub>2</sub> per kWh; tree equivalence at 45 trees per tonne is illustrative, not measured sequestration.

BASIS AND LIMITATIONS

This is an indicative financial analysis for a standard 500 kWp commercial rooftop plant. **It is not a quotation, an offer, or tax advice.** Pricing is subject to a site survey, roof condition, structure height, cable routing, evacuation distance and the APSPDCL sanction actually granted.

Tariffs, the depreciation block and the corporate tax rate are as we understand them at the date printed below and are subject to change by APERC and by the Finance Act. The tax treatment shown assumes a GST-registered company that has opted into s.115BAA and has sufficient other business income to absorb the early-year depreciation. **Have your chartered accountant confirm the tax position before you rely on it.**

Generation depends on site irradiance, shading, orientation, soiling and grid availability. Actual results will differ from this projection.